

INFO 5602
Information Visualization

“The Importance of Context”

Agenda

- Effective data visualization *does/should not* start with *data* or *data visualization*
- Exploratory vs. explanatory analysis
- Four questions for defining context: ***why?, who?, what?, how?***
- Consulting for context
- Storyboarding

Effective data visualization *does/should not* start with data or data visualization

- Effective data visualization/communication should **always start** with focus on/ paying sufficient attention to understand the **context** for the need to communicate
- The context (or environment) for communication shapes the overall structure of the relationship between you (information transmitter) and your stakeholders/audience (information recipients)
- Context is the starting point before you define any problem you want to solve with visualization or machine learning modelling
- Poor understanding of context can damage **effective stakeholder communication**

Exploratory vs. explanatory analysis

- Explanatory analysis = exploratory analysis + **context**
- Exploratory analysis is what you do to understand the data and find insights to highlight to your stakeholders
- Explanatory analysis involves **using context to telling compelling stories** about the insights you uncover from exploratory analysis
- **Context** tells you:
 - **Why** you need to communicate (with your particular stakeholder(s))
 - **Who** you are communicating with + your relationship with your stakeholders
 - **What** you are communicating - i.e what data are you going to use
 - **How** you intend to communicate what you want to present
- Always focus on **explanatory analysis** rather than exploratory analysis to achieve effective stakeholder communication

Context: *why?*

- Always start any decision making with **why**; ask yourself and your stakeholders: why?
- **Why?** guides you on the following:
 - A definition of or motivation or the need to communicate.
 - **Should you communicate at all? Doing nothing is always an option.**
 - **Has the stakeholder asked for communication?**
 - **If communication has the potential to add quantifiable benefits to your stakeholders relative to doing nothing, you should probably push to communicate with your audience.**

Context: *who?*

- Once you are clear on your “**why?**”, you need to address “**who?**”.
- **Who? = Your audience + You**
- **Who?** guides you on the following:
 - To whom are you communicating? Who are the crucial decision makers?
 - How does your stakeholder perceive you and what you are?
 - Do you have a (good) relationship with your stakeholders?
 - It is important to have a good understanding of who your audience is and **how they perceive you**. This can help you identify common ground that will help you ensure your stakeholders hear your message?
 - To what **extent does your stakeholder want to change?**
 - To what extent does your stakeholder want you to tell them what they want to hear?
 - Can your stakeholder handle the truth?

Your audience

- The **more specific you can be about who your audience is**, the better position you will be in for effective stakeholder communication
- You should understand the **preferences of your stakeholders to the extent you can**. Gather information about them so you can adjust your storytelling on the fly and **tailor your presentation to their preferences**
- Identifying key decision makers is one way of **narrowing your audience**
- The **more you know about your audience**, the better positioned you will be to understand how to **resonate with them and achieve effective stakeholder communication**
- Remember: communication is not about you; its about your audience

You

- It is helpful to think about the relationship you have with your stakeholders/audiences and how they perceive you; **perception is reality for many people**
- Will you be meeting your stakeholders/audiences **for the first time** through your visualization presentation or do you have an established relationship?
- Do your stakeholders already trust you as an expert or do you need to work to establish your credibility?
- Your relationship with your stakeholders/audience determines how presentation structuring and order and flow of the overall story you tell

Context: *what?*

- Once you are clear on your “**why?**” and “**what**”, you need to address “**what?**”.
- ***What?* = Action + Mechanism + Tone**
- **What?** guides you on the following:
 - What do you need your stakeholders to know or do (**action**)? What have they asked for?
 - Making your communication relevant to the needs/requirements of your stakeholders?
 - Form a clear understanding of why your stakeholders should care about what you have to present to them?
 - Level of detail (control **mechanism**) needed in your presentation
 - Communication **tone** you adopt at the time of presentation

Mechanism

LIVE PRESENTATION WRITTEN DOC OR EMAIL

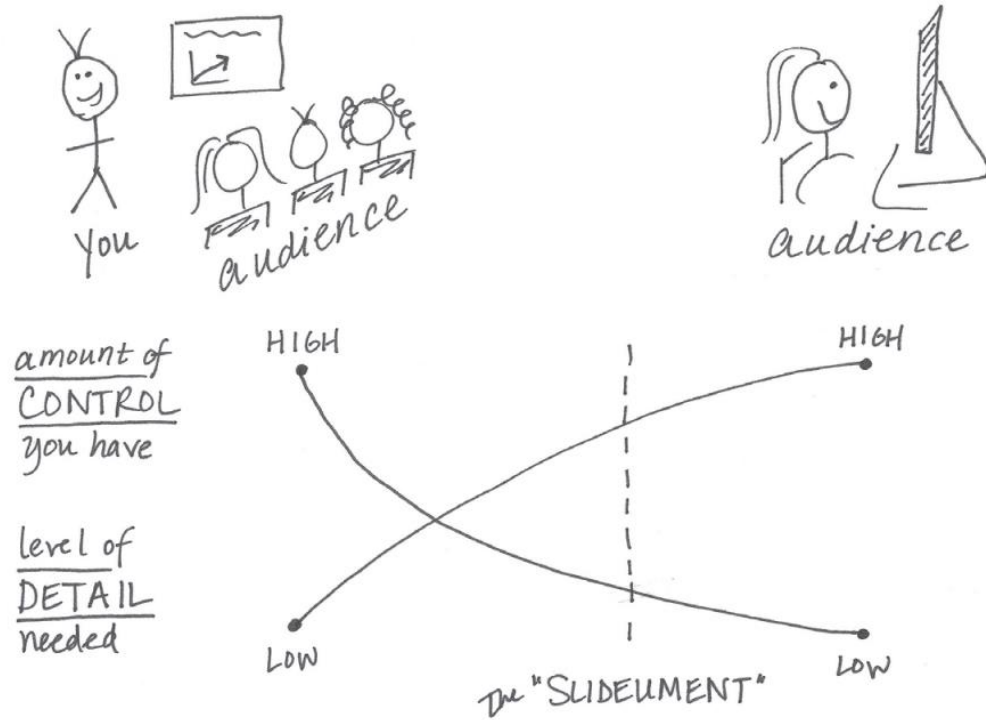


FIGURE 1.1 Communication mechanism continuum

Source: Knaflic (2015)

- Mechanism refers to the **method** you will communicate to your audience
- The method you use to communicate is determined by:
 - **Amount of control** you will have over how the audience takes in information
 - **Level of detail** that is needed for presented information
- With live presentation you have **high control** over how the audience takes in information; so, you need a **low level of detail** in your presentation
- With non-live presentation (e.g. email) you have **low control** over how the audience takes in information; so, you need a **high level of detail in your presentation**

Tone

- It is helpful to think about the tone you want your communication to set given the context and your peculiar stakeholder needs
- What tone do you want your communication to convey to your audience?
 - Urgency
 - Deliberative
 - Light
- The tone you desire for your audience has **implications for the visualization and communication design** choices you make
- Whatever tone you decide for your visualizations has to use the principles of **user-centered design (UCD)**

Context: *How?*

- Only when you are clear on your “**why?**”, “**who?**” and “**what?**” do you need to address “**How?**”.
- ***What?* = Action + Mechanism + Tone**
- **How?** guides you on the following:
 - **What data are available** that will help make the point your stakeholders need?
 - Data (and their visualization) is the supporting evidence of the story you present
 - It might be useful to **consult with your key decision makers** about excluding **non-supporting information** before presenting to a wider group of stakeholders
 - Non-supporting information adds little value and may **increase cognitive load**

Consulting for Context : Questions to ask

- You might need to consult with your stakeholders to fully understand the context around your data visualization and presentation
- **Questions to ask your stakeholder include:**
 - What background information is relevant or essential?
 - What biases do our audience have that might make them supportive of or resistant to our visual presentation?
 - What does the audience already know (redundant information) that you need to de-emphasize in favor of information they actually care about?
 - What data is available that would strengthen reception? Is the audience familiar with the data or are they new?
 - What would a successful visual presentation look like?
 - If you could use a single sentence to summarize the message, what would it be?
- Boiling the message down to a (few) sentence(s) might help with repetition (storytelling)

Storyboarding

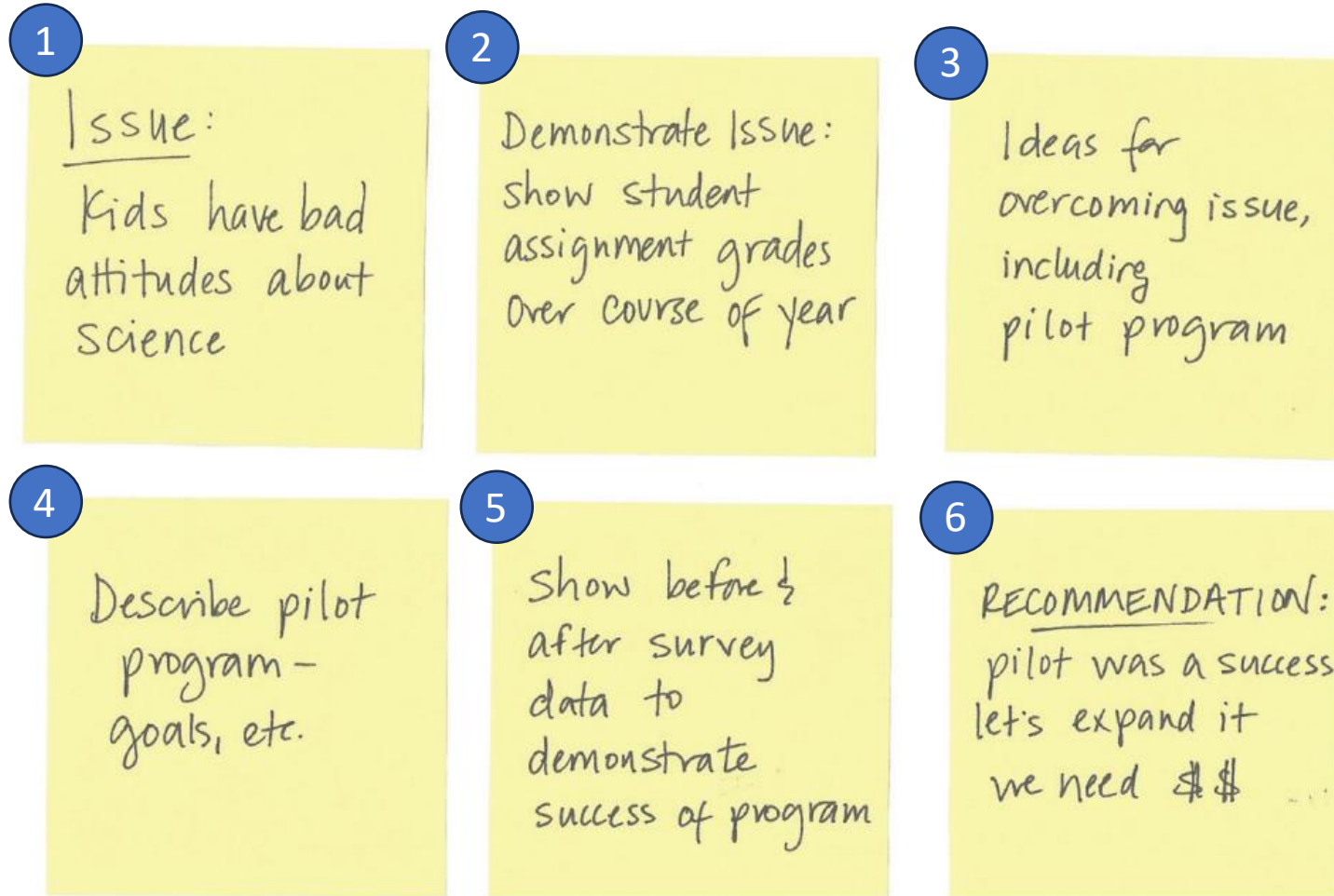


FIGURE 1.2 Example storyboard

Source: Knaflic (2015)

- **Storyboarding** is what you do upfront to ensure your visualization communication is effective with your stakeholders.
- A **storyboard** establishes visual **presentation structuring** – it's a visual outline of the content you plan to create and
- When it comes to visualization/storyboarding, **do not start with presentation software** (Excel, Tableau etc).
 - Keeps you off “slide-generating mode”; makes you to think about how the pieces of your story fit together
 - Use low tech (e.g. paper, post-it notes) to construct your storyboard

End of presentation